3. If the velocity of an object increases with respect to time, then the object is said to be in_____ acceleration.
4. The slope of the speed–time graph gives _____.
5. In _____ equilibrium, the centre of gravity remains at the same height when it is displaced.

III. Match the following.

Displacement	Knot
Light travelling through vacuum	Geometric centre
Speed of ship	Metre
Centre of gravity of geometrical shaped objects	Larger base area
Stability	Uniform velocity

IV. Analogy

1. Velocity : metre/second :: Acceleration : _____ .
2. Length of scale : metre :: Speed of aeroplane : _____ .
3. Displacement / Time : Velocity :: Speed / Time : _____ .

V. Answer very briefly.

1. Asher says all objects having uniform speed need not have uniform velocity. Give reason.
2. Saphira moves at a constant speed in the same direction. Rephrase the same sentence in fewer words using concepts related to motion.
3. Correct your friend who says that acceleration gives the idea of how fast the position changes.

VI. Answer briefly.

1. Show the shape of the distance – time graph for the motion in the following cases.
   a. A bus moving with a constant speed.
   b. A car parked on a road side.
2. Distinguish between speed and velocity.
3. What do you mean by constant acceleration?
4. What is centre of gravity ?

VII. Answer in detail.

1. Explain the types of stability with suitable examples.
2. Write about the experiment to find the centre of gravity of the irregularly shaped plate.

VIII. Numerical problems.

1. Geetha takes 15 minutes from her house to reach her school on a bicycle. If the bicycle has a speed of 2 m/s, calculate the distance between her house and the school.
2. A car starts from rest and it is travelling with a velocity of 20 m/s in 10 s. What is its acceleration?
3. A bus can accelerate with an acceleration of 1 m/s^2. Find the minimum time for the bus to attain the speed of 100 km/s from 50 km/s.

IX. Fill in the boxes.

S.No.	First Move	Seconde Move	Distance (m)	Displacement
1.	Move 4 metres east	Move 2 metres west	6	2 m east
2.	Move 4 metres north	Move 2 metres south		
3.	Move 2 metres east	Move 4 metres west		
4.	Move 5 metres east	Move 5 metres west		
5.	Move 5 metres south	Move 2 metres north		
6.	Move 10 metres west	Move 3 metres east		

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